

Cooking Companion

Open it, say what you want to make, and it writes the recipe, checks what you have, and talks you through it a step at a time — interrupting, remembering and timing the way a person in the kitchen with you would.

github.com/chemtrailx/forg3-3
Next.js 16.3 · React 19.1 · TypeScript 5.9
Supabase Postgres · Groq · Rime
Deployed in **bom1** · beside the database

12

SERVER TOOLS

278

TESTS PASSING

87

SOURCE MODULES

7

RUNTIME DEPS

3

DB MIGRATIONS

~150ms

AUTH ROUND TRIP

ONE CONVERSATIONAL TURN

CAPTURE

The microphone is open all session. A pure-policy detector adapts its floor to the room and uses hysteresis — less energy is needed to stay in speech than to enter it — so a quiet syllable never ends a sentence.

ENDPOINT

Transcription starts at a **420 ms** pause, inside an **1100 ms** hangover. Resume and the work is discarded; stop and it was the whole utterance. Low latency without truncating anyone.

REASON

One cancellable SSE request carries state, tool calls and PCM together, so a barge-in is a single abort reaching model, tools and synthesiser at once.

SPEAK

The first pass is streamed and spoken clause by clause; a tool slower than **250 ms** earns a filler, so there is no dead air.

Heard-transcript ledger

Rime's word timestamps, matched against samples the speaker actually played, give the word an interruption landed on. That — not the sentence it meant to say — is what history records, so the model never answers a correction using words the cook never heard.

DATA & ISOLATION

IDENTITY

Supabase Auth. Nothing downstream takes a user id as an argument; every query is built from the verified session cookie.

DATABASE

Row-level security **enabled and forced** on every user table, with ownership and new-account triggers.

BELT & BRACES

Every read also filters by user_id, so a policy regression is caught by a query rather than by a user.

TABLES

profiles · recipes · cooking_sessions · timers · user_memory · conversation_turns

Proven, not asserted

The suite runs against an in-memory PostgREST double that simulates RLS, so an isolation failure is a wrong row in a test rather than another cook's recipe on screen.

PROVIDERS & RESILIENCE

SPEECH OUT

Rime **coda / astra**, PCM 24 kHz on a pooled ws3 socket. operation:clear cancels at the source; contextIds make stale audio an equality check.

SPEECH IN

Whisper large-v3-turbo, primed with the dish and ingredients in play — the words a general model most often gets wrong.

REASONING

Groq **gpt-oss-20b, 120b** as its rate-limit fallback, a second vendor behind that. Free tiers throughout.

FAILURE

A limit is waited out or answered elsewhere; failover never restarts an answer already being heard.

Provider-agnostic

Each vendor sits behind one interface — TtsProvider, SttProvider, LlmProvider. Swapping one changes a file.

